



NVIDIA Multi-Instance GPU (MIG)

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MIG

- Multi-Instance GPU (MIG) expands the performance and value of NVIDIA Blackwell and Hopper generation GPUs.
- MIG can partition the GPU into as many as seven instances
 - Each fully isolated with its own high-bandwidth memory, cache, and compute cores.
- This gives administrators the ability to support small to large workload with guaranteed quality of service (QoS)
- Each MIG instance behaves like a standalone GPU to applications, so there's no change to the CUDA platform.

Without MIG and with MIG

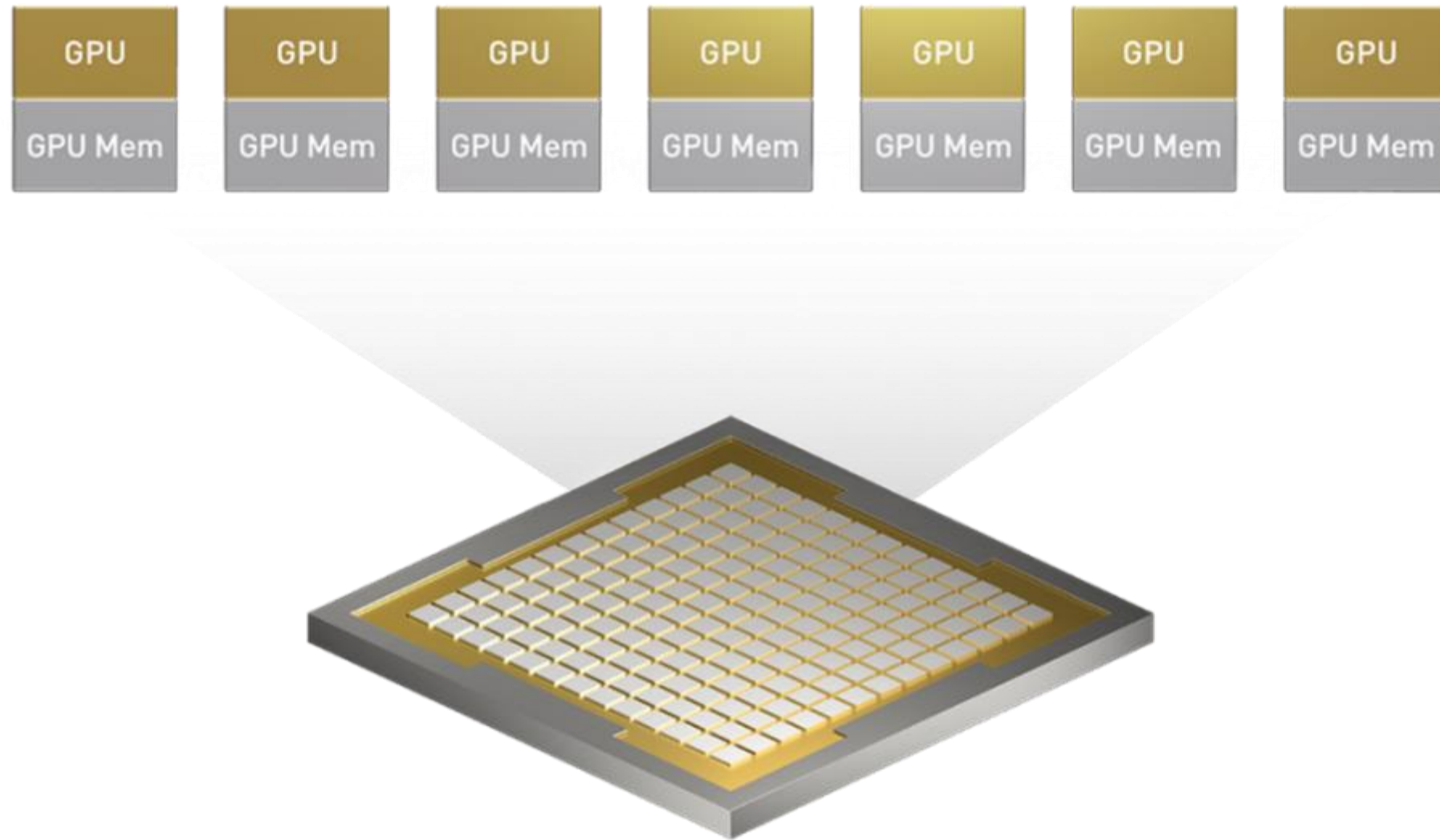
Without MIG, different jobs running on the same GPU, such as different AI inference requests, compete for the same resources.

A job consuming larger memory bandwidth starves others,

→ several jobs miss their latency targets.

With MIG, jobs run simultaneously on different instances, each with dedicated resources for compute, memory, and memory bandwidth,

It results in predictable performance with QoS and maximum GPU utilization.



	Blackwell Ultra GPU	Blackwell GPU	H100 GPU
Confidential computing	Yes	Yes	Yes
Instance types	Up to 7x 34GB Up to 4x 70GB Up to 2x 140GB Up to 1x 288 GB	Up to 7x 23GB Up to 4x 45GB Up to 2x 95GB Up to 1x 192GB	7x 10GB 4x 20GB 2x 40GB 1x 80GB
GPU profiling and monitoring	Concurrently on all instances	Concurrently on all instances	Concurrently on all instances
Secure Tenants	7x	7x	7x

Benefits

MIG provides the flexibility to choose many different instance sizes, which allows provisioning of the right-sized GPU instance for each workload

MIG enables inference, training, and high-performance computing (HPC) workloads to run at the same time on a single GPU with deterministic latency and throughput.

Unlike time slicing, each workload runs in parallel, delivering higher performance.

Dedicated video decoders for each MIG instance deliver secure, high-throughput intelligent video analytics (IVA) on shared infrastructure.

With concurrent MIG profiling, administrators can monitor right-sized GPU acceleration and allocate resources for multiple users.

For researchers with smaller workloads, rather than renting a full cloud instance, they can use MIG to isolate a portion of a GPU securely while being assured that their data is secure at rest, in transit, and in use.

Reconfigurability

MIG instances can also be dynamically reconfigured, enabling administrators to shift GPU resources in response to changing user and business demands.

For example, seven MIG instances can be used **during the day** for low-throughput inference and reconfigured to one large MIG instance **at night** for deep learning training.